

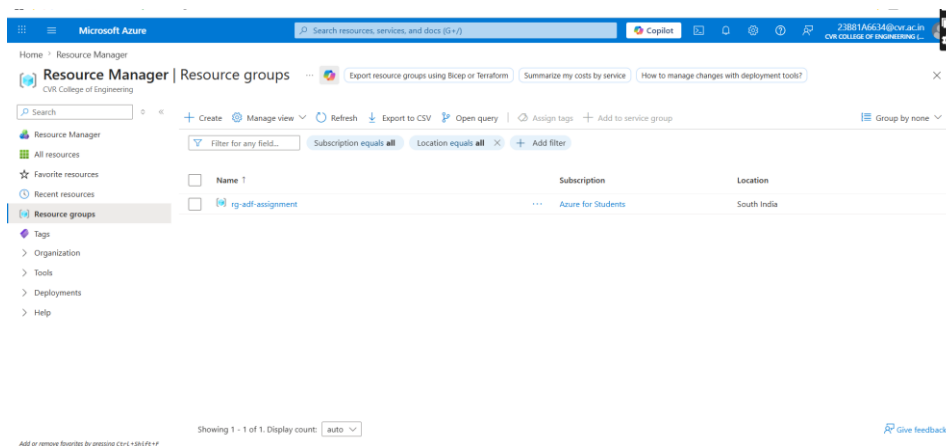
Week 4 Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



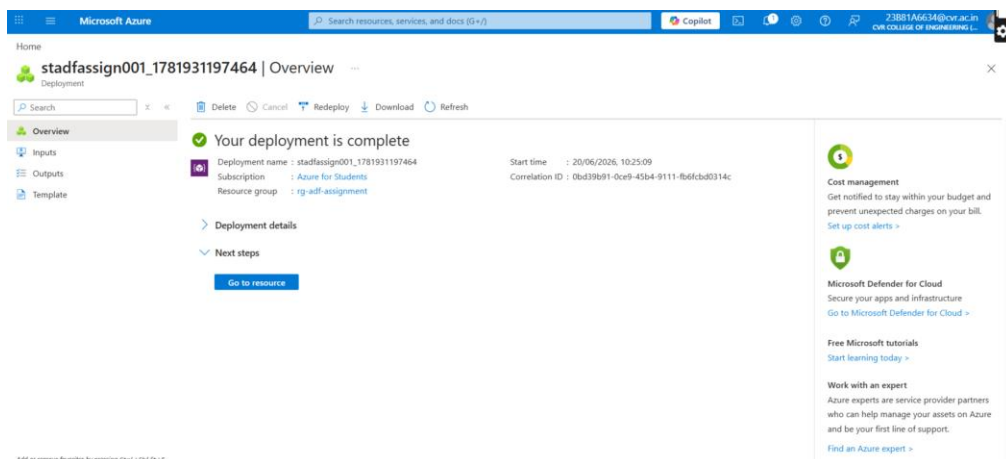
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



Microsoft Azure

Home > Storage center > Blob Storage

Generate a guide for using Azure storage accounts | Analyze alerts across storage accounts | Generate cost-saving LCM rules for my storage accounts

Search

Summary Resources

Overview

All storage resources

Object storage

File storage

Block storage

Data management

Migration

Partner solutions

Management services

Help

+ Create + Restore Manage view Refresh Export to CSV Open query Assign tags Delete Add to service group Group by none

Filter for any field... Subscription equals all Resource Group equals all Location equals all Add filter

Name	Type	Kind	Resource Group	Location	Subscription
stadfassign001	Storage account	StorageV2	rg-adf-assignment	East Asia	Azure for Students

Showing 1 - 1 of 1. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Microsoft Azure

Home > Storage center | Blob Storage > stadfassign001

stadfassign001 | Containers

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Add or remove favorites by pressing Ctrl+Shift+F

+ Add container Upload Refresh Delete

Search containers by prefix Only show active containers

Showing all 2 items

access level	Lease state	Storage connector
	Available	-
	Available	-

Microsoft Azure

Home > Storage center | Blob Storage > stadfassign001 | Containers

source-container

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

source-container

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Sample - Superstore.csv	6/20/2026, 10:31:58 AM	Hot (Inferred)	Block blob	2.18 MiB	Available

Add or remove favorites by pressing Ctrl+Shift+F

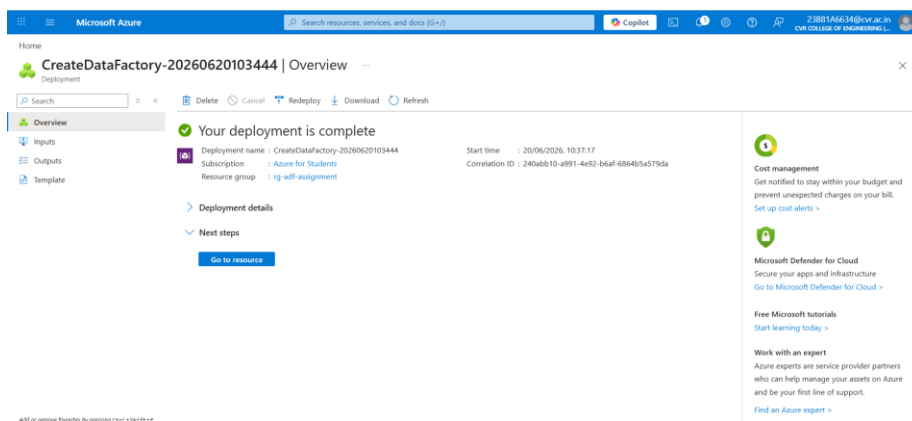
Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity



Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

Microsoft Azure | Data Factory | adf-assignment-demo

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Validate all Publish all

Migrate to Fabric (Preview)

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations: Any

Showing 1 - 1 of 1 items

Name	Type	Related	Annotations
LS_BlobStorage	Azure Blob Storage	0	

Microsoft Azure | Data Factory | adf-assignment-demo

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Validate all Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - DS_SourceCSV
- Data flows 0
- Power Query 0

DS_SourceCSV

DelimitedText
DS_SourceCSV

Connection Schema Parameters

Linked service * LS_BlobStorage

File path * source-container / Directory / Sample - Superstore.csv

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\n, or \r\n)

Encoding Default(UTF-8)

Quote character Double quote (")

Properties

General Related

Name * DS_SourceCSV

Description

Annotations + New

Cancel

Microsoft Azure | Data Factory | adf-assignment-demo

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Validate all Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 2
 - DS_SourceCSV
 - DS_DestinationCSV
- Data flows 0
- Power Query 0

DS_DestinationCSV

DelimitedText
DS_DestinationCSV

Connection Schema Parameters

Linked service * LS_BlobStorage

File path * destination-container / Directory / File name

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\n, or \r\n)

Encoding Default(UTF-8)

Properties

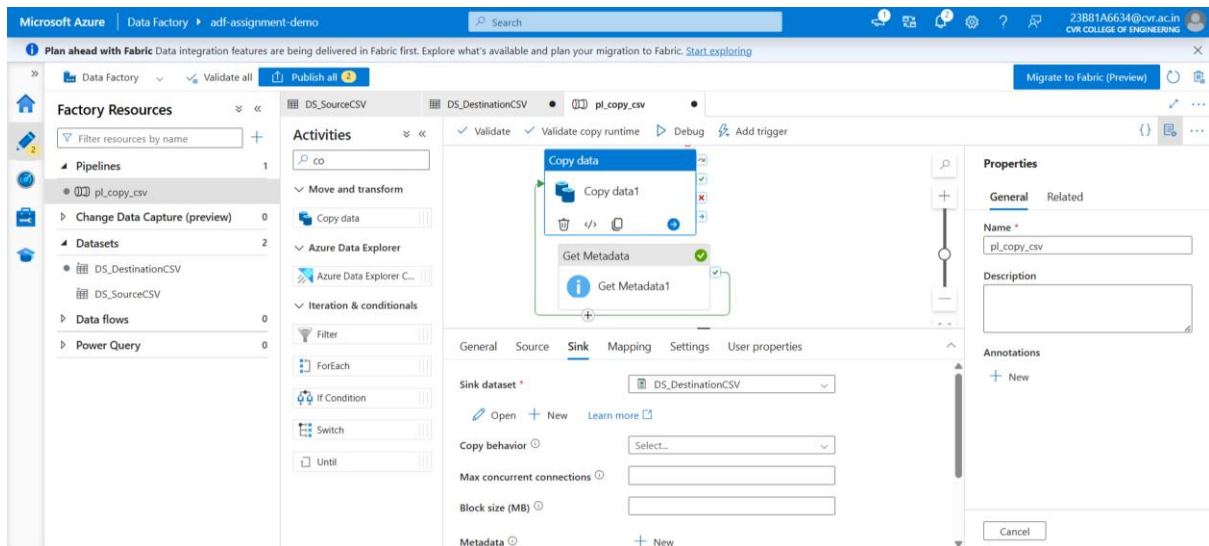
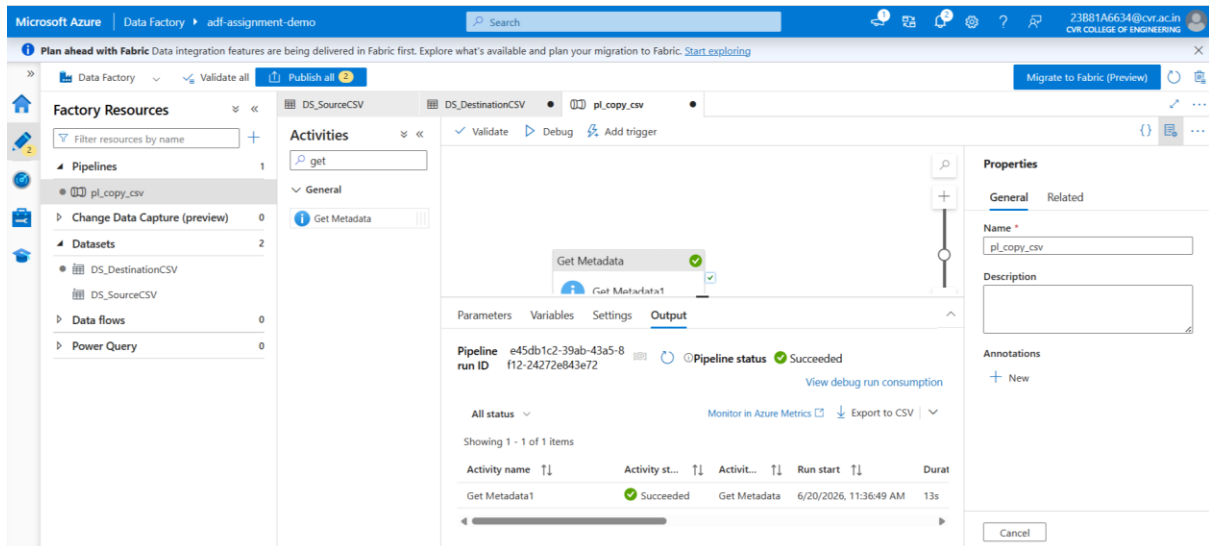
General Related

Name * DS_DestinationCSV

Description

Annotations + New

Cancel



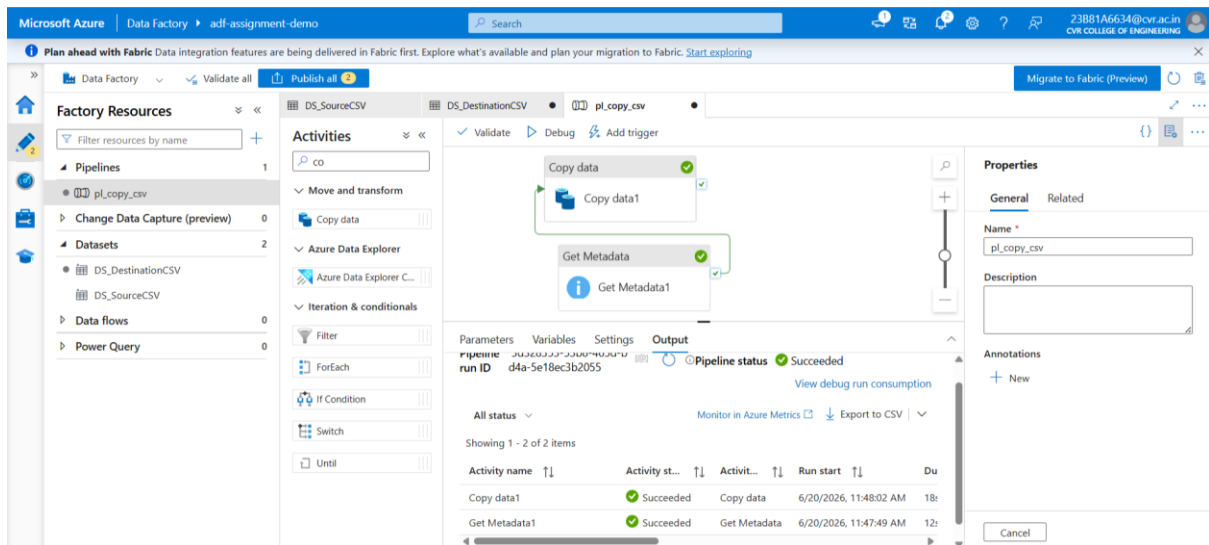
Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



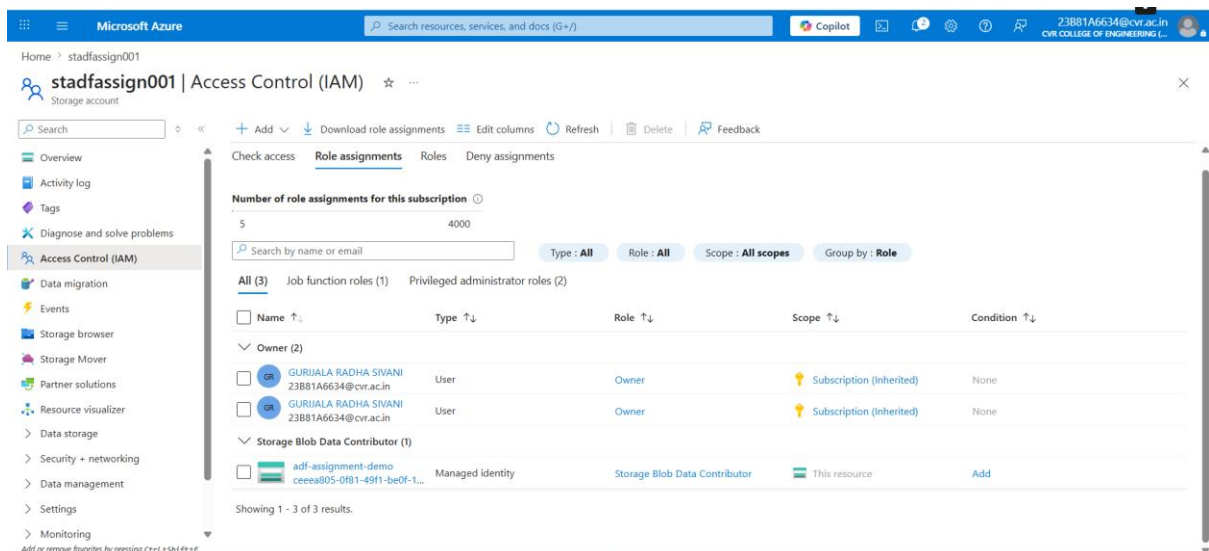
Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

Ans:

Mini Project Requirement:	Already done in
Source: CSV file in Blob:-	Task 2
Linked Service + Dataset + Pipeline:-	Task 3 & 4
Process: Copy Data + Metadata check :-	Task 4 (Get Metadata → Copy Data)
Destination: New file/location:-	Task 4/5 (destination-container)
Pipeline executed successfully:-	Task 5
Data copied to destination:-	Task 5 (you verified the file landed there)
Metadata validated:-	Task 3 (Get Metadata output JSON)

Note:

- ➔ I wasn't able to find the plain **Contributor** role in the role-assignment search; it kept showing only specialized roles like "API Management Contributor." Since my account already has **Owner** access (which includes Contributor and Reader permissions), I proceeded with that instead. I did complete the key part — assigning **Storage Blob Data Contributor** to the ADF managed identity for storage access.
- ➔ I have also displayed the summary of weekly assignments in the readme in my github Repository.

➔ Summary:-

Azure ADF Project — Quick Summary

- **Task 1:** Created Resource Group rg-adf-assignment
- **Task 2:** Created Storage Account stadfassign001, uploaded CSV to source-container, made destination-container
- **Task 3:** Created ADF adf-assignment-demo, set up Linked Service + source/destination datasets, ran Get Metadata activity successfully
- **Task 4:** Built pipeline pl_copy_csv (Get Metadata → Copy Data)
- **Task 5:** Ran pipeline via Debug — **Succeeded**, file confirmed in destination
- **Task 6:** Owner role already covers Reader/Contributor; assigned **Storage Blob Data Contributor** to ADF's managed identity for storage access
- **Mini Project:** Fully satisfied by the same pipeline — CSV copied end-to-end with metadata validated